DFS + backtracking search tree: Rotating Workforce Scheduling

DFS calls 51 · tried 119 · rejected 69 · backtracked 22 · solution depth 28

d=0

start

d=1

W1 Mon = D

d=2

W2 Mon = D

d=3

W3 Mon = D ×
Mon needs 2 D

W3 Mon = N

d=4

W4 Mon = D ×
Mon needs 2 D

W4 Mon = N ×
Mon needs 1 N

W4 Mon = O

d=5

W1 Tue = D

d=6

W2 Tue = D

d=7

W3 Tue = D ×
Tue needs 2 D

W3 Tue = N

d=8

W4 Tue = D ×
Tue needs 2 D

W4 Tue = N ×
Tue needs 1 N

W4 Tue = O

d=9

W1 Wed = D

d=10

W2 Wed = D

d=11

W3 Wed = D ×
Wed needs 2 D

W3 Wed = N

d=12

W4 Wed = D ×
Wed needs 2 D

W4 Wed = N ×
Wed needs 1 N

W4 Wed = O

d=13

W1 Thu = D

d=14

W2 Thu = D

d=15

W3 Thu = D ×
Thu needs 2 D

W3 Thu = N
exhausted: 3 nodes

W3 Thu = O

d=16

W4 Thu = D ×
Thu needs 2 D

W4 Thu = N

d=17

W1 Fri = D

d=18

W2 Fri = D
exhausted: 39 nodes

W2 Fri = N
exhausted: 9 nodes

W2 Fri = O

d=19

W3 Fri = D

d=20

W4 Fri = D ×
Fri needs 2 D

W4 Fri = N

d=21

W1 Sat = D ×
work block > 5

W1 Sat = N ×
work block > 5

W1 Sat = O

d=22

W2 Sat = D
exhausted: 6 nodes

W2 Sat = N
exhausted: 6 nodes

W2 Sat = O

d=23

W3 Sat = D

d=24

W4 Sat = D ×
Sat needs 1 D

W4 Sat = N

d=25

W1 Sun = D

d=26

W2 Sun = D ×
Sun needs 1 D

W2 Sun = N
exhausted: 3 nodes

W2 Sun = O

d=27

W3 Sun = D ×
Sun needs 1 D

W3 Sun = N

d=28

W4 Sun = D ×
Sun needs 1 D

W4 Sun = N ×
Sun needs 1 N

W4 Sun = O
GOAL

Solution path · Goal · Invalid × · Exhausted subtree